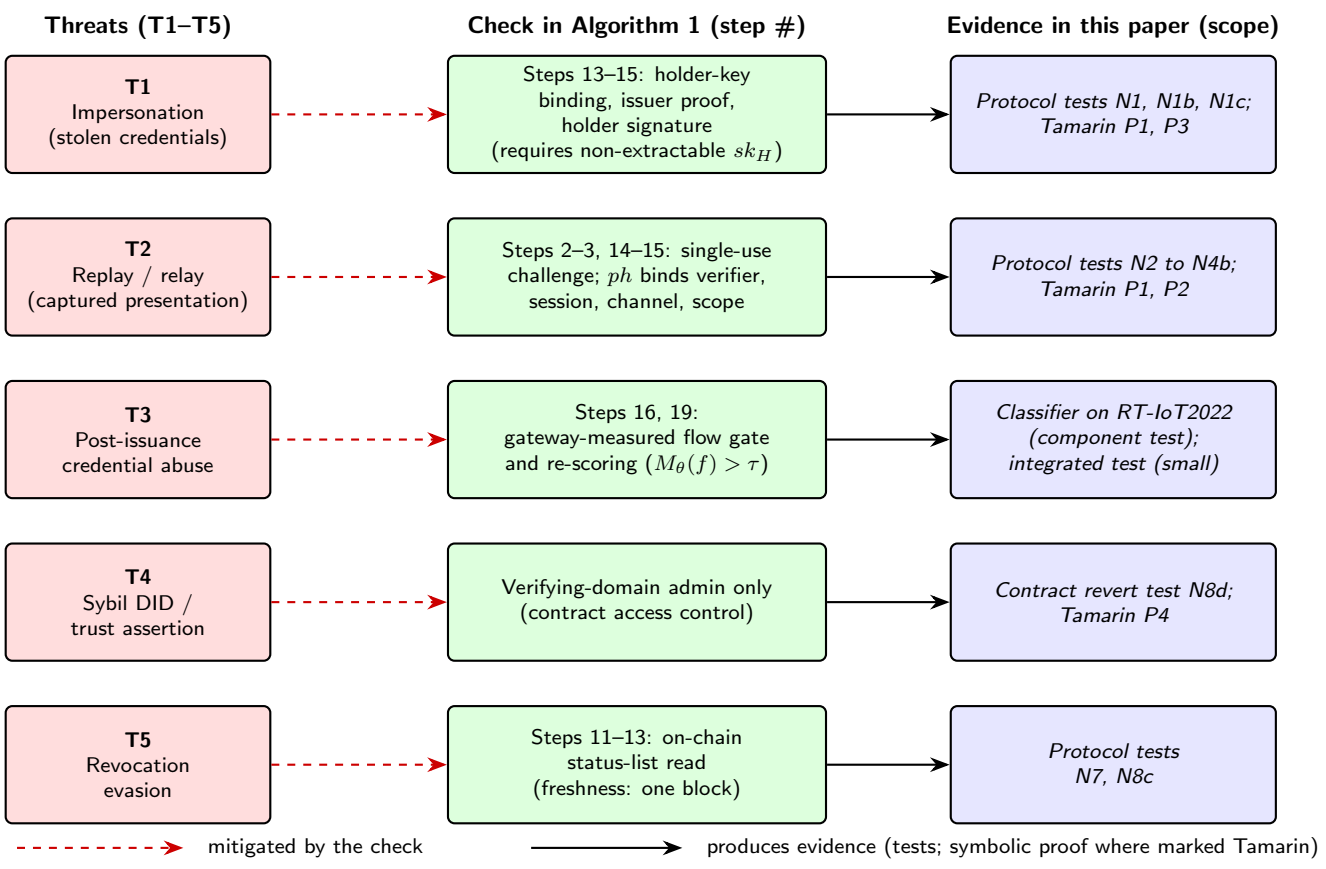

Threats (T1–T5)
Check in Algorithm 1 (step #)
Evidence in this paper (scope)
T1
Impersonation
(stolen credentials)
Steps 13–15: holder-key
binding, issuer proof,
holder signature
(requires non-extractable sk_H)
Protocol tests N1, N1b, N1c;
Tamarin P1, P3
T2
Replay / relay
(captured presentation)
Steps 2–3, 14–15: single-use
challenge; ph binds verifier,
session, channel, scope
Protocol tests N2 to N4b;
Tamarin P1, P2
T3
Post-issuance
credential abuse
Steps 16, 19:
gateway-measured flow gate
and re-scoring ($M_\theta(f) > \tau$)
Classifier on RT-IoT2022
(component test);
integrated test (small)
T4
Sybil DID /
trust assertion
Verifying-domain admin only
(contract access control)
Contract revert test N8d;
Tamarin P4
T5
Revocation
evasion
Steps 11–13: on-chain
status-list read
(freshness: one block)
Protocol tests
N7, N8c
mitigated by the check
produces evidence (tests; symbolic proof where marked Tamarin)